Intraocular Pressure (IOP) and Aqueous Chambers

Images: 6.2, 6.10, 6.3, 6.5, 6.8, 6.9, 6.10

I. Three ocular chambers

A. Vitreous chamber

B. Posterior chamber

C. Anterior chamber

1. Boundaries

a. Anteriorly:

b. Posteriorly:

c. Peripherally:

2. Anterior chamber angle

    a. Aqueous humor exits the anterior chamber.

    b. Scleral spur

        (1) Dense collagen fibers

        (2) Function: Attachment for ciliary muscle fibers

    c. Trabecular meshwork

        (1) Two anatomic divisions

            i. Corneoscleral meshwork

            ii. Uveal meshwork

d. Schlemm canal

(1) Circular vessel that allows aqueous to pass out of the anterior chamber.

(2) External collector channels empty into deep scleral or intrascleral plexus and then go to the episcleral or conjunctival veins

3. Gonioscopy appearance

a. Gonioscopy uses a special lens that can overcome the total internal reflection. The examiner can then view the structures of the anterior chamber angle.

II. Aqueous dynamics

A. Function of aqueous:

B. Aqueous production:

C. Aqueous outflow

1. Conventional outflow: trabecular flow

2. Unconventional outflow: uveoscleral flow